

6.4.1 多元复合函数微分法

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一、复合函数的中间变量均为一元函数

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所以 $\Delta z = \frac{\partial z}{\partial u} \Delta u + \frac{\partial z}{\partial v} \Delta v + o(\rho)$, 其中 $\rho = \sqrt{(\Delta u)^2 + (\Delta v)^2}$.

于是 $\frac{\Delta z}{\Delta x} = \frac{\partial z}{\partial u} \frac{\Delta u}{\Delta x} + \frac{\partial z}{\partial v} \frac{\Delta v}{\Delta x} + \frac{o(\rho)}{\Delta x}.$

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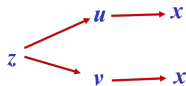
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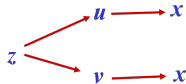


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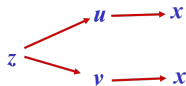


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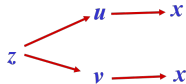
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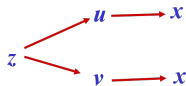
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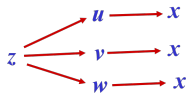
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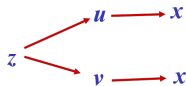


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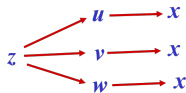


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设 $z = f(u, v)$, 而 $u = u(x, y)$, $v = v(x, y)$, 若 $u = u(x, y)$, $v = v(x, y)$ 在点 (x, y) 处偏导数都存在, 且 $z = f(u, v)$ 在相应点 (u, v) 可微,

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设 $z = f(u, v)$, 而 $u = u(x, y)$, $v = v(x, y)$, 若 $u = u(x, y)$, $v = v(x, y)$ 在点 (x, y) 处偏导数都存在, 且 $z = f(u, v)$ 在相应点 (u, v) 可微, 则复合函数 $z = f(u(x, y), v(x, y))$ 在点 (x, y) 处存在偏导数, 且

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存在偏导数, 且

$$\begin{cases} \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} \\ \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} \end{cases}$$

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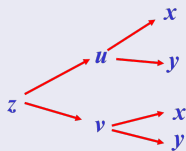
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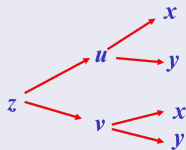
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链式法则: 按线相乘, 分线相加.

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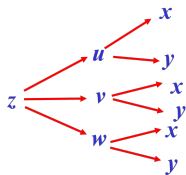
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注意: 在复合函数的求导过程中, 如果出现某一函数的中间变量是一元函数, 则涉及它的偏导数的记号应改为一元函数的导数记号.

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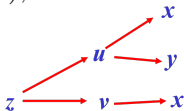
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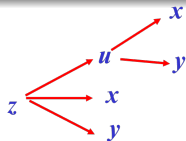
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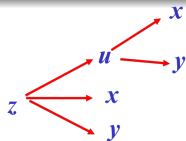


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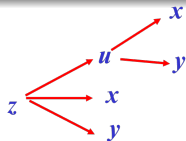
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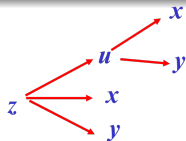
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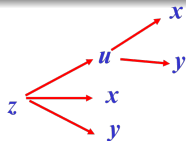
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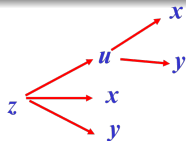
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其中 f_i ($i = 1, 2, 3$) 或 φ_i ($i = 1, 2$) 表示函数 f 或 φ 对第 i 个变量的偏导数.

例3. 设 $z = xy + xF(u)$, 而 $u = \frac{y}{x}$, $F(u)$ 为可导函数, 证明:

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = z + xy.$$

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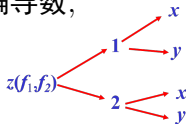
例4. 设 $z = f(x - y, xy^2)$, f 有二阶连续偏导数,

$$\text{求 } \frac{\partial z}{\partial x}, \frac{\partial^2 z}{\partial x^2}, \frac{\partial^2 z}{\partial x \partial y}.$$

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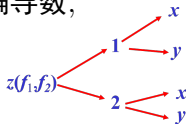
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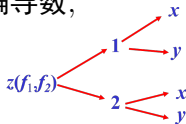


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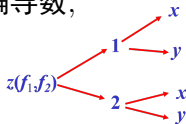
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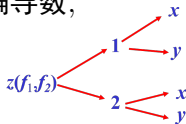
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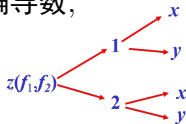
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例5. 设 $z = f(xy, \frac{x}{y}) + g(\frac{y}{x})$, f 有二阶连续偏导数, g 有二阶连续导数, 求 $\frac{\partial^2 z}{\partial x \partial y}$.

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例6. 设 $u = u(\xi, \eta)$ 具有二阶连续偏导数, 其中 $\xi = x + ay$,
 $\eta = x + by$ ($a \neq b$), 问 a, b 为何值时, 可使
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$$\begin{aligned} \therefore u_{xx} + 4u_{xy} + 3u_{yy} &= (3a^2 + 4a + 1)u_{\xi\xi} + (6ab + 4a + 4b + 2)u_{\xi\eta} \\ &\quad + (3b^2 + 4b + 1)u_{\eta\eta}, \end{aligned}$$

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解得 $a = -\frac{1}{3}, b = -1$ 或 $a = -1, b = -\frac{1}{3}$.

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例7. 设 $u = f(x, y, z)$, $y = \varphi(x, t)$, $t = \psi(x, z)$ 均可微,
求 $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial z}$.

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6.4.4 隐函数微分法

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定义 (隐函数)

设有方程 $F(x_1, x_2, \cdots, x_n, y) = 0$, 如果存在一个 n 元函数

$y = \varphi(x_1, x_2, \cdots, x_n)$ ($(x_1, x_2, \cdots, x_n) \in \Omega \subseteq \mathbb{R}^n$, Ω 为一区域)

使得将 $y = \varphi(x_1, x_2, \cdots, x_n)$

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$$F(x_1, x_2, \dots, x_n, \varphi(x_1, x_2, \dots, x_n)) \equiv 0,$$

则称 $y = \varphi(x)$ 是由方程 $F(x_1, x_2, \dots, x_n, y) = 0$ 确定的**隐函数**.

定理6.4.4 (隐函数存在定理)

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- (1) 在点 (x_0, y_0) 的某一邻域内具有连续偏导数 F_x, F_y ;
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则方程 $F(x, y) = 0$ 在 x_0 的某邻域 $N(x_0, \delta)$ 内唯一确定了一个具有连续导数的函数 $y = f(x)$, 它满足 $y_0 = f(x_0)$ 及 $F(x, f(x)) \equiv 0$ ($x \in N(x_0, \delta)$), 并有

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则方程 $F(x, y) = 0$ 在 x_0 的某邻域 $N(x_0, \delta)$ 内唯一确定了一个具有连续导数的函数 $y = f(x)$, 它满足 $y_0 = f(x_0)$ 及

$F(x, f(x)) \equiv 0$ ($x \in N(x_0, \delta)$), 并有 $\frac{dy}{dx} = -\frac{F_x}{F_y}$.

例8. 验证方程 $x^2 + y^2 - 1 = 0$ 在点 $(0, 1)$ 的某邻域内能唯一确定一个具有连续导数、且当 $x = 0$ 时 $y = 1$ 的隐函数 $y = f(x)$, 并求此函数的一阶导数在 $x = 0$ 的值.

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则 $F_x = 2x$, $F_y = 2y$, $F(0, 1) = 0$, $F_y(0, 1) = 2 \neq 0$,

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故由定理6.4.4可知, 方程 $x^2 + y^2 - 1 = 0$ 在点 $(0, 1)$ 的某邻域内

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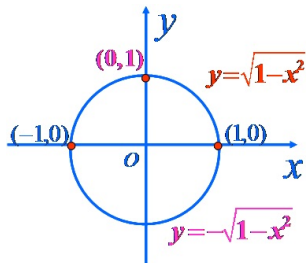
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$$\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{2x}{2y} = -\frac{x}{y}, \quad \left. \frac{dy}{dx} \right|_{x=0} = 0.$$

方程 $x^2 + y^2 - 1 = 0$ 表示单位圆.

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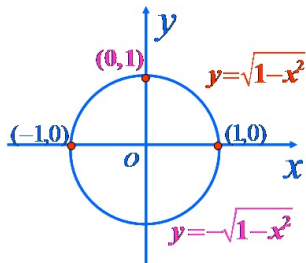
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附近的一段圆弧曲线的方程就可用

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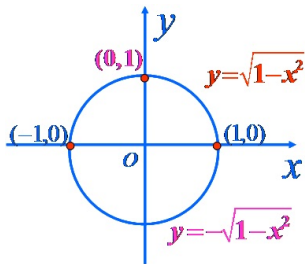
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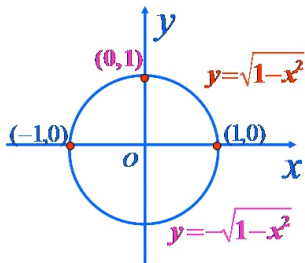
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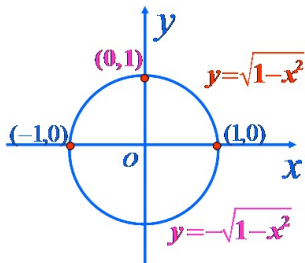
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由此可体会定理6.4.4中条件 $F_y(x_0, y_0) \neq 0$ 的意义.



例9. 求由方程 $\ln \sqrt{x^2 + y^2} = \arctan \frac{y}{x}$ 所确定的隐函数

$y = y(x)$ 的导数 $\frac{dy}{dx}$.

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解: 设 $F(x, y) = \ln \sqrt{x^2 + y^2} - \arctan \frac{y}{x}$, 则

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$$F_x = \frac{x}{x^2 + y^2} - \frac{-\frac{y}{x^2}}{1 + (\frac{y}{x})^2} = \frac{x + y}{x^2 + y^2},$$

$$F_y = \frac{y}{x^2 + y^2} - \frac{-\frac{1}{x}}{1 + (\frac{y}{x})^2} = \frac{y - x}{x^2 + y^2},$$

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定理6.4.5

设 $n + 1$ 元函数 $F(x_1, \cdots, x_n, y)$ 满足下列条件:

- (1) 在点 $(x_1^0, \cdots, x_n^0, y_0)$ 的某邻域内具有连续的偏导数;
- (2) $F(x_1^0, \cdots, x_n^0, y_0) = 0$;
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则方程 $F(x_1, \cdots, x_n, y) = 0$ 在 $M_0(x_1^0, \cdots, x_n^0)$ 的某邻域 $N(M_0, \delta)$ 内唯一确定了一个具有连续偏导数的函数 $y = f(x_1, \cdots, x_n)$, 它满足 $y_0 = f(x_1^0, \cdots, x_n^0)$ 及 $F(x_1, \cdots, x_n, f(x_1, \cdots, x_n)) \equiv 0$ ($(x_1, \cdots, x_n) \in N(M_0, \delta)$), 并有

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$$\frac{\partial y}{\partial x_i} = -\frac{F_{x_i}}{F_y}, \quad i = 1, \cdots, n.$$

定理

设三元函数 $F(x, y, z)$ 满足下列条件:

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- (2) $F(x_0, y_0, z_0) = 0$;
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$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}, \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$$

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解法2：公式法

$$\text{令 } F(x, y, z) = x^2 + 2y^2 + 3z^2 - 4,$$

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解法3：利用一阶微分形式不变性 $x^2 + 2y^2 + 3z^2 = 4$,

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$$\therefore \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right)$$

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$$\therefore \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right) = -\frac{x}{3} \cdot \left(-\frac{1}{z^2} \right) \cdot \frac{\partial z}{\partial y}$$

解法2：公式法

$$\text{令 } F(x, y, z) = x^2 + 2y^2 + 3z^2 - 4,$$

$$\text{则 } F_x = 2x, F_y = 4y, F_z = 6z,$$

$$\therefore \frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{x}{3z}, \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z} = -\frac{2y}{3z}.$$

解法3：利用一阶微分形式不变性 $x^2 + 2y^2 + 3z^2 = 4$, 微分得:

$$2xdx + 4ydy + 6zdz = 0, \quad \therefore dz = -\frac{x}{3z}dx - \frac{2y}{3z}dy.$$

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例11. 设 $z = z(x, y)$ 是由方程 $F(x - az, y - bz) = 0$ 所确定的隐函数, 其中 a, b 为常数, F 具有一阶连续偏导数, 证明:

$$a \frac{\partial z}{\partial x} + b \frac{\partial z}{\partial y} = 1.$$

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$$\text{故 } a \frac{\partial z}{\partial x} + b \frac{\partial z}{\partial y} = \frac{aF_1}{aF_1 + bF_2} + \frac{bF_2}{aF_1 + bF_2} = 1.$$

例12. 设 $u = f(x, y, z) = e^x y z^2$, 其中 $z = z(x, y)$ 是由方程 $x + y + z + xyz = 0$ 所确定的隐函数, 求 $du|_{(0,1)}$.

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$$\text{则 } \frac{\partial u}{\partial x} \Big|_{(0,1)} = 1, \quad \frac{\partial u}{\partial y} \Big|_{(0,1)} = 3, \quad \text{故 } du|_{(0,1)} = dx + 3dy.$$

问: $\begin{cases} y = y(x) \\ z = z(x) \end{cases}$ 由方程组 $\begin{cases} F(x, y, z) = 0 \\ G(x, y, z) = 0 \end{cases}$ 所确定, 其中 F, G 可微,

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解上述方程组可得 $\frac{dy}{dx}$ 和 $\frac{dz}{dx}$:

$$\frac{dy}{dx} = \frac{\begin{vmatrix} -F_x & F_z \\ -G_x & G_z \end{vmatrix}}{\begin{vmatrix} F_y & F_z \\ G_y & G_z \end{vmatrix}}, \quad \frac{dz}{dx} = \frac{\begin{vmatrix} F_y & -F_x \\ G_y & -G_x \end{vmatrix}}{\begin{vmatrix} F_y & F_z \\ G_y & G_z \end{vmatrix}},$$

$$\frac{dy}{dx} = \frac{\begin{vmatrix} -F_x & F_z \\ -G_x & G_z \end{vmatrix}}{\begin{vmatrix} F_y & F_z \\ G_y & G_z \end{vmatrix}},$$

$$= -\frac{1}{J} \frac{\partial(F, G)}{\partial(x, z)},$$

$$\frac{dz}{dx} = \frac{\begin{vmatrix} F_y & -F_x \\ G_y & -G_x \end{vmatrix}}{\begin{vmatrix} F_y & F_z \\ G_y & G_z \end{vmatrix}}$$

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其中, $J = \frac{\partial(F, G)}{\partial(y, z)}.$

由方程组确定的隐函数

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定理6.4.6

设三元函数 $F(x, y, z), G(x, y, z)$ 满足下列条件:

- (1) 在点 $M_0(x_0, y_0, z_0)$ 的某个邻域内有对各个变量的连续偏导数;
- (2) $F(x_0, y_0, z_0) = G(x_0, y_0, z_0) = 0$;
- (3) Jacobi行列式 $J = \frac{\partial(F, G)}{\partial(y, z)}$ 满足

$$J|_{M_0} = \frac{\partial(F, G)}{\partial(y, z)} \Big|_{M_0} = \begin{vmatrix} F_y & F_z \\ G_y & G_z \end{vmatrix} \Big|_{M_0} \neq 0,$$

则由方程组 $\begin{cases} F(x, y, z) = 0 \\ G(x, y, z) = 0 \end{cases}$ 在 x_0 的某邻域 $N(x_0, \delta)$ 内确定

了唯一的一组具有连续导数的函数 $\begin{cases} y = y(x) \\ z = z(x) \end{cases}$, 它们满足

$$\begin{cases} y_0 = y(x_0) \\ z_0 = z(x_0) \end{cases} \quad \text{及} \quad \begin{cases} F(x, y(x), z(x)) \equiv 0, \\ G(x, y(x), z(x)) \equiv 0, \end{cases} \quad x \in N(x_0, \delta), \text{ 并且}$$

$$\frac{dy}{dx} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(x, z)}, \quad \frac{dz}{dx} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(y, x)}.$$

定理5.5 (隐函数存在定理)

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设有函数方程组
$$\begin{cases} F(x, y, u, v) = 0 \\ G(x, y, u, v) = 0 \end{cases},$$

定理5.5 (隐函数存在定理)

设有函数方程组
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且四元函数 $F(x, y, u, v), G(x, y, u, v)$ 满足下列条件:

- (1) 在 $M_0(x_0, y_0, u_0, v_0)$ 的某个邻域内有连续偏导数;
- (2) $F(x_0, y_0, u_0, v_0) = G(x_0, y_0, u_0, v_0) = 0$;
- (3) Jacobi 行列式

$$J|_{M_0} = \frac{\partial(F, G)}{\partial(u, v)} \Big|_{M_0} = \begin{vmatrix} F_u & F_v \\ G_u & G_v \end{vmatrix} \Big|_{M_0} \neq 0,$$

则由方程组
$$\begin{cases} F(x, y, u, v) = 0 \\ G(x, y, u, v) = 0 \end{cases}$$
 在点 (x_0, y_0) 的某邻域 $N((x_0, y_0), \delta)$ 内

确定了唯一的一组具有连续导数的函数
$$\begin{cases} u = u(x, y) \\ v = v(x, y) \end{cases},$$

它们满足
$$\begin{cases} u_0 = u(x_0, y_0) \\ v_0 = v(x_0, y_0) \end{cases} \quad \text{及} \quad \begin{cases} F(x, y, u(x, y), v(x, y)) \equiv 0, \\ G(x, y, u(x, y), v(x, y)) \equiv 0, \end{cases}$$

 $(x, y) \in N((x_0, y_0), \delta)$, 并且

$$\frac{\partial u}{\partial x} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(x, v)}, \quad \frac{\partial v}{\partial x} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(u, x)},$$

$$\frac{\partial u}{\partial y} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(y, v)}, \quad \frac{\partial v}{\partial y} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(u, y)}.$$

例5. 求出方程组 $\begin{cases} xu - yv = 0 \\ yu + xv = 1 \end{cases}$ 所确定的隐函数的偏导数

$$\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}.$$

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解法1: 将方程组两边对 x 求导, 得

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解法1: 将方程组两边对 x 求导, 得

$$\begin{cases} u + x \frac{\partial u}{\partial x} - y \frac{\partial v}{\partial x} = 0 \\ y \frac{\partial u}{\partial x} + v + x \frac{\partial v}{\partial x} = 0 \end{cases}$$

例5. 求出方程组 $\begin{cases} xu - yv = 0 \\ yu + xv = 1 \end{cases}$ 所确定的隐函数的偏导数

$$\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}.$$

解法1: 将方程组两边对 x 求导, 得

$$\begin{cases} u + x \frac{\partial u}{\partial x} - y \frac{\partial v}{\partial x} = 0 \\ y \frac{\partial u}{\partial x} + v + x \frac{\partial v}{\partial x} = 0 \end{cases} \Rightarrow \begin{cases} x \frac{\partial u}{\partial x} - y \frac{\partial v}{\partial x} = -u \\ y \frac{\partial u}{\partial x} + x \frac{\partial v}{\partial x} = -v \end{cases}, \text{求解可得}$$

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$$\frac{\partial u}{\partial x} = \frac{\begin{vmatrix} -u & -y \\ -v & x \end{vmatrix}}{\begin{vmatrix} x & -y \\ y & x \end{vmatrix}} = -\frac{xu + yv}{x^2 + y^2}, \quad \frac{\partial v}{\partial x} = \frac{\begin{vmatrix} x & -u \\ y & -v \end{vmatrix}}{\begin{vmatrix} x & -y \\ y & x \end{vmatrix}} = \frac{yu - xv}{x^2 + y^2}.$$

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将 du, dv 视为未知变量, 解得

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例6. 设函数 $u = u(x)$ 由方程组
$$\begin{cases} u = f(x, y) \\ g(x, y, z) = 0 \\ h(x, z) = 0 \end{cases}$$
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其中 f, g, h 可微, 且 $\frac{\partial h}{\partial z} \neq 0, \frac{\partial g}{\partial y} \neq 0$, 求 $\frac{du}{dx}$.

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故 $\frac{du}{dx} = f_x + f_y \frac{dy}{dx} = f_x + f_y \frac{h_x g_z - g_x h_z}{h_z g_y}$.

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$$\therefore \frac{du}{dx} = f_x + f_y \frac{h_x g_z - g_x h_z}{h_z g_y}.$$